

Title: Smart Energy Monitoring System for Reducing Electricity Consumption in School Facilities

Background of the Study

Electricity consumption is a major operating cost for many schools and universities. Classrooms, laboratories, offices, libraries, and common areas use lights, air conditioning units, computers, projectors, and other electrical equipment. In some cases, energy is wasted because devices are left on even when rooms are not in use. Manual monitoring of electricity usage can be difficult because staff cannot always check every room or facility.

A smart energy monitoring system can help track electricity consumption in real time. It can collect data from sensors or meters and display energy usage through a dashboard. This type of system may help facility managers identify high-consumption areas, detect unusual usage, and promote energy-saving behavior. For schools, reducing electricity consumption can lower operational costs and support environmental sustainability.

This study aims to design and assess a smart energy monitoring system for school facilities. The system will focus on monitoring electricity usage and providing information that can help administrators make better energy management decisions.

Statement of the Problem

The main problem addressed by this study is the lack of real-time visibility of electricity consumption in school facilities. Without accurate and timely data, it is difficult to identify where energy is being wasted. This may lead to higher electricity costs and inefficient use of resources.

Research Objectives

This study aims to:

1. Identify common sources of electricity consumption in school facilities.
2. Design a smart energy monitoring system for selected rooms or areas.
3. Measure electricity usage through sensors, meters, or simulated monitoring data.
4. Provide a dashboard that shows energy usage patterns.
5. Recommend energy-saving actions based on the collected data.

Proposed Methodology

This study will use a design and development research approach. The proposed system may include sensors, microcontrollers, energy meters, and a web-based dashboard. If actual hardware implementation is limited, simulated energy data may be used for the MVP.

The system will collect or simulate data such as room location, device type, usage time, energy consumption, and estimated cost. The data will be displayed in charts or summary reports. The system will also identify rooms or devices with high electricity usage.

Expected Output

The expected output of this study is a working prototype or simulated system that monitors electricity consumption in school facilities. The system should display energy usage, identify high-consumption areas, and provide basic recommendations for reducing electricity waste.

Significance of the Study

This study may benefit school administrators, facility managers, and maintenance staff by giving them better visibility of electricity usage. It may also help schools reduce operating costs and promote responsible energy consumption. Students and employees may also become more aware of the importance of energy conservation.

Keywords

Smart energy monitoring, electricity consumption, school facilities, energy efficiency, sensors, dashboard, sustainability, facility management, real-time monitoring, energy conservation

SAMPLE CONCEPT PAPER 7

Statistics Topic

Title: Predictive Analysis of Student Retention Using Academic Performance and Enrollment Data

Background of the Study

Student retention is an important concern for schools and universities. When students stop enrolling or delay their studies, it affects academic planning, institutional performance, and student success. Schools often collect data such as grades, enrollment status, attendance, program, academic standing, and completion progress. However, these data are not always used to predict which students may be at risk of dropping out or becoming inactive.

Predictive analysis can help identify patterns in student data. By using academic performance and enrollment records, schools may detect early warning signs of retention problems. For example, low grades, repeated subjects, delayed enrollment, or incomplete requirements may indicate that a student needs academic support. With this information, administrators can provide interventions earlier.

This study aims to use predictive analysis to examine student retention based on academic performance and enrollment data. The study will focus on identifying variables that may help predict whether students are likely to continue, delay, or stop their studies.

Statement of the Problem

The main problem addressed by this study is the limited use of existing student data for predicting retention risks. Schools may already have academic and enrollment records, but these records are often used only for documentation. This study seeks to determine how statistical analysis can help identify students who may be at risk of not continuing their studies.

Research Objectives

This study aims to:

1. Identify academic and enrollment variables related to student retention.
2. Analyze patterns in student performance and enrollment behavior.
3. Develop a simple predictive model for identifying students at risk.
4. Evaluate the accuracy or usefulness of the predictive model.
5. Recommend ways to use retention data for student support and academic planning.

Proposed Methodology

This study will use a quantitative research design. The data may include student grades, enrollment history, program, academic status, number of completed units, failed subjects, and term-to-term enrollment records. If real data is not available, sample or simulated student records may be used for system testing.

Statistical methods such as frequency analysis, correlation, logistic regression, or classification may be used depending on the available data. The study may also compare students who continued enrollment with those who became inactive or delayed.

Expected Output

The expected output of this study is a predictive analysis report or prototype model that identifies students who may be at risk of retention problems. The output may include risk categories such as low risk, moderate risk, and high risk. The study may also provide a dashboard or summary table for academic staff.

Significance of the Study

This study may benefit schools by helping them make better use of student data. It may support early intervention, academic advising, and retention planning. Students may also benefit if the school can identify concerns earlier and provide support before problems become more serious.

Keywords

Predictive analysis, student retention, academic performance, enrollment data, risk prediction, statistical modeling, logistic regression, student success, early intervention, education analytics